D-MAVT
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Analysis III

Prof. A. Iozzi
ETH Zürich
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Solutions Serie 8

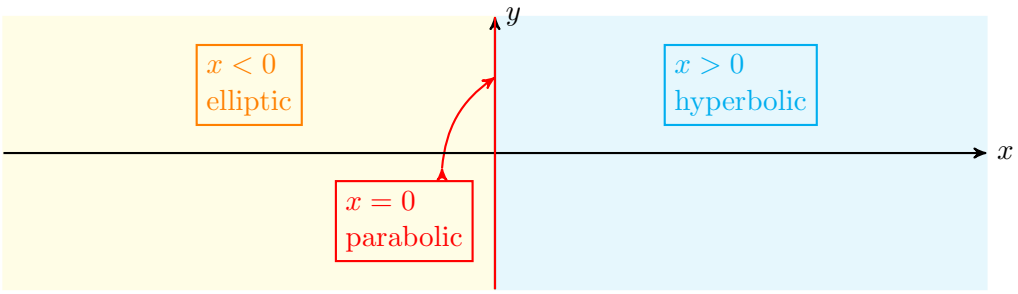
Theory reminder on the classification of PDEs: A 2nd order PDE is an equation of the form:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} = F(x, y, u, u_x, u_y)$$

where the coefficients A, B, C may also be functions of x, y. We say that the PDE is, respectively, hyperbolic, parabolic or elliptic, if the function $AC - B^2$ is, respectively, always smaller, equal, or greater than zero. When the sign changes in different regions of the plane (x, y), the equation is called *of mixed type*.
For example the Euler-Tricomi equation

$$u_{xx} - xu_{yy} = 0$$

has $AC - B^2 = 1 \cdot (-x) - (0)^2 = -x$, and therefore is of mixed type: hyperbolic in the half plane $x > 0$, elliptic in the other half plane $x < 0$, and parabolic on the line $x = 0$.





Solutions Serie 8

Please turn!

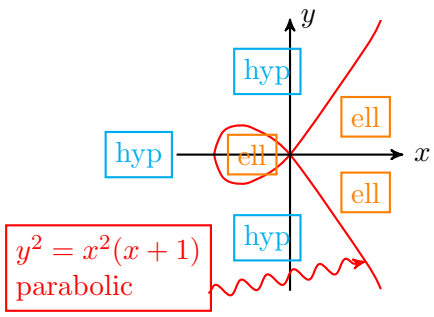
1. Consider the following PDEs - in what follows, $u = u(x, y)$ is a function of two variables.

$$u_{xx} + 2u_{xy} + u_{yy} + 3u_x + xu = 0, \tag{1}$$
$$u_{xx} + 2u_{xy} + 2u_{yy} + u_y = 0, \tag{2}$$
$$u_{xx} + 8u_{xy} + 2u_{yy} + e^x u_x = 0, \tag{3}$$
$$yu_{xx} + 2xu_{xy} + u_{yy} - u_y = 0, \tag{4}$$
$$(x+1)u_{xx} + 2yu_{xy} + x^2 u_{yy} = 0. \tag{5}$$

Which of this is hyperbolic? Parabolic? Elliptic? Of mixed type?
In the last case, try to understand in which region of the plane (x, y) they are hyperbolic, parabolic or elliptic[1].

Solution:

(1) $AC - B^2 = 1 - 1 = 0 \quad \rightsquigarrow \quad$ parabolic.
(2) $AC - B^2 = 2 - 1 = 1 > 0 \quad \rightsquigarrow \quad$ elliptic.
(3) $AC - B^2 = 2 - 4^2 = -14 < 0 \quad \rightsquigarrow \quad$ hyperbolic.
(4) $AC - B^2 = y - x^2 \quad \rightsquigarrow \quad$ of mixed type. The equation $y - x^2 = 0$ describes a parabola in the plane. Above the parabola the equation is elliptic, on the parabola it is parabolic, and below it is hyperbolic.
(5) $AC - B^2 = (x+1)x^2 - y^2 \quad \rightsquigarrow \quad$ of mixed type. The equation $(x+1)x^2 - y^2 = 0$ describes the following curve.



The closed region in the middle and the right region is where $(x+1)x^2 - y^2 > 0$ and therefore the PDE is elliptic. The big open region remaining is where the PDE is hyperbolic, while the red curve describes the points in which it is parabolic.

2. Consider the following functions.

a) $u(x, t) = e^{-100t}\cos(2x)$

b) $u(x, t) = \sin(2x)\cos(8t)$

c) $u(x, t) = e^{-36t}\sin(3x)$

[1] You can plot the curve $\{AC - B^2 = 0\}$ on - say - Wolfram|Alpha to understand its shape.

Look at the next page!

Which PDE between the heat equation, $u_t = c^2 u_{xx}$, and the wave equation, $u_{tt} = c^2 u_{xx}$, does each of these solve? Write down also which is the constant c in each case.

Solution:

a) $u(x,t) = e^{-100t}\cos(2x)$.

Solution:

We have

$$\begin{cases} u_t = -100e^{-100t}\cos(2x) = -100u \\ u_{tt} = \cdots = 100^2 u \\ u_{xx} = \cdots = -4u \end{cases}$$

To answer the question we are asked to compare either the first, or the second term, with a positive multiple of the third. This is possible (only) for the first term, and we need coefficient $c = 5$. Therefore u is a solution of the heat equation:

$$u_t = -100u = 5^2(-4u) = c^2 u_{xx}.$$

b) $u(x,t) = \sin(2x)\cos(8t)$.

Solution:

Here

$$\begin{cases} u_t = -8\sin(2x)\sin(8t) \\ u_{tt} = \partial_t(-8\sin(2x)\sin(8t)) = -64\sin(2x)\cos(8t) = -64u \\ u_{xx} = \cdots = -4u \end{cases}$$

In this case instead u is a solution of the wave equation, with coefficient $c = 4$:

$$u_{tt} = -64u = 4^2(-4u) = c^2 u_{xx}.$$

c) $u(x,t) = e^{-36t}\sin(3x)$.

Solution:

This is analogous to the first one.

$$\begin{cases} u_t = -36e^{-36t}\sin(3x) = -36u \\ u_{tt} = \cdots = 36^2 u \\ u_{xx} = \cdots = -9u \end{cases}$$

so we have a solution of the heat equation, this time with coefficient $c = 2$

$$u_t = -36u = 2^2(-9u) = c^2 u_{xx}.$$

3. Find the general solution $u = u(x, y)$ for the following PDEs:

a) $u_y + 2yu = 0$

Solution:

In this equation there are no derivatives in the variable x anywhere. We can thus think of this as a family of ODEs, one for each $x \in \mathbb{R}$. More formally for each fixed x the function $u(x, \cdot)$ must be a solution of

$$u_y + 2yu = 0.$$

We have already met equations of this kind and the general solution is given by some constant factor multiplied by $e^{-\int 2y} = e^{-y^2}$. Except that this time the 'constant' factor may depend on x, and is not constant anymore. The general solution is

$$u(x, y) = f(x)e^{-y^2},$$

where $f(x)$ is *any* function.

b) $u_{yy} = 4xu_y$.

Solution:

This is somehow similar because there are still no derivatives on x. First we call $v := u_y$ and we obtain the equation

$$v_y = 4xv.$$

As before this can be solved by solving the correspondent problem for $v(x, \cdot)$ for each fixed x. This time the general solution is

$$v(x, y) = f(x)e^{4xy},$$

and by integrating in y we get general solution

$$u(x, y) = g(x) + \frac{f(x)}{4x}e^{4xy}.$$

4. Fourier transform (*8 Points*)

Compute the Fourier transform of the following function for $\omega \neq 0$

$$f(x) = \begin{cases} \sqrt{2\pi}(1 + x), & 0 \leq x \leq \pi, \\ 0, & \text{otherwise.} \end{cases}$$

Solution:

Recall that the Fourier transform of f is given by

$$\mathcal{F}(f)(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{-iwx}\, dx.$$

Therefore, for $\omega \neq 0$,

$$\begin{aligned}
\mathcal{F}(f)(w) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{0}^{\pi} \sqrt{2\pi}(1+x) e^{-iwx}\, dx \\
&= \int_{0}^{\pi} e^{-iwx}\, dx + \int_{0}^{\pi} x e^{-iwx}\, dx \\
&= \frac{e^{-iwx}}{-iw}\bigg|_0^{\pi} + \left[x \frac{e^{-iwx}}{-iw} \right]_0^{\pi} - \int_0^{\pi} \frac{e^{-iwx}}{-iw}\, dx \\
&= \frac{1}{-iw}(e^{-iw\pi} - 1) + \pi \frac{\mathrm{e}^{-iw\pi}}{-iw} - \frac{e^{-iwx}}{(-iw)^2}\bigg|_0^{\pi} \\
&= \frac{1}{-iw}(e^{-iw\pi} - 1) + \pi \frac{\mathrm{e}^{-iw\pi}}{-iw} + \frac{e^{-iw\pi}}{w^2} - \frac{1}{w^2} \\
&= \left(\frac{1}{-iw} + \frac{\pi}{-iw} + \frac{1}{w^2} \right) e^{-iw\pi} + \frac{1}{iw} - \frac{1}{w^2} \\
&= \left(\frac{i}{w} + \frac{i\pi}{w} + \frac{1}{w^2} \right) e^{-iw\pi} - \frac{i}{w} - \frac{1}{w^2}.
\end{aligned}$$

And for for $\omega = 0$, we have

$$\begin{aligned}
\mathcal{F}(f)(0) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{0}^{\pi} \sqrt{2\pi}(1+x)\, dx \\
&= \pi + \frac{\pi^2}{2}.
\end{aligned}$$

5. Let $a > 0$ be a positive number. Recall that the Fourier transform of the Gaussian function with parameter a is:

$$\mathcal{F}(\mathrm{e}^{-ax^2})(\omega) = \frac{1}{\sqrt{2a}} \mathrm{e}^{-\frac{\omega^2}{4a}}$$

Use this equality to calculate the following integrals:

a) $\displaystyle\int_{\mathbb{R}} \mathrm{e}^{-ax^2} dx$

b) $\displaystyle\int_{\mathbb{R}} x\mathrm{e}^{-ax^2} dx$

c) $\displaystyle\int_{\mathbb{R}} x^2\mathrm{e}^{-ax^2} dx$

Hint: Use the properties of the Fourier transform, from §3.6 of the Lecture Notes.

Solution:

By the definition of the Fourier transform of a function:

$$\widehat{g}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(x) \mathrm{e}^{-i\omega x} dx \,,$$

if we know $\widehat{g}(\omega)$ then we recover the integral of the function as

$$\int_{\mathbb{R}} g(x) dx = \sqrt{2\pi}\, \widehat{g}(0) \,.$$

In light of this, in order to solve the exercise we need the Fourier transform of $\mathrm{e}^{-ax^2}, x\mathrm{e}^{-ax^2}, x^2\mathrm{e}^{-ax^2}$. The last two can be found from the first one by computing the iterated derivatives of the Fourier transform:

$$\widehat{(x^k f)} = i^k \frac{d^k}{d\omega^k} \widehat{f} \,.$$

In conclusion in general we have

$$\int_{\mathbb{R}} x^k f(x) dx = \sqrt{2\pi}\, i^k \left(\frac{d^k}{d\omega^k} \widehat{f} \right) (0) \,.$$

In the following we use $f(x) = \mathrm{e}^{-ax^2}$ and we compute the iterated derivatives of its Fourier transform, evaluated in zero.

a)

$$\int_{\mathbb{R}} \mathrm{e}^{-ax^2} = \sqrt{2\pi} \frac{1}{\sqrt{2a}} = \sqrt{\frac{\pi}{a}} \,.$$

b)

$$\frac{d}{d\omega} \widehat{f} = \frac{1}{\sqrt{2a}} \mathrm{e}^{-\frac{\omega^2}{4a}} \left(-\frac{2\omega}{4a} \right) \quad \Longrightarrow \quad \int_{\mathbb{R}} x\mathrm{e}^{-ax^2} dx = 0$$

(this could also simply be deduced from the fact that the function in the integral is odd, without using any Fourier transform).

c)

$$\frac{d^2}{d\omega^2} \widehat{f} = \frac{1}{\sqrt{2a}} \mathrm{e}^{-\frac{\omega^2}{4a}} \left(\frac{\omega^2}{4a^2} - \frac{1}{2a} \right) \quad \Longrightarrow \quad \int_{\mathbb{R}} x^2 \mathrm{e}^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \frac{1}{2a} \,.$$